

WaveState: A parallel selective state-space attention model with class-adaptive time-frequency features for multivariate time series classification[★]

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ARTICLE INFO

Keywords:

Multivariate time series
Classification
Class-adaptive wavelet transform
Selective state-space model
Sparse neural network

ABSTRACT

Multivariate time series classification (MTSC) is crucial for handling complex dynamic systems and enhancing decision-making across various fields. Although current deep learning-based approaches have made significant progress in MTSC, they still face limitations in effectively capturing subtle categorical distinctions. Meanwhile, balancing feature discrimination with computational efficiency remains a challenge, particularly when working with complex, high-dimensional time series data. To address these issues, we propose a novel MTSC approach, WaveState, based on our parallel selective state-space attention framework, combined with the class-adaptive wavelet transform. This approach allows for the flexible decomposition of multi-scale frequency domain features, thereby enhancing the extraction of subtle inter-class differences. The multi-scale parallel convolutional structure, working in conjunction with the parallel selective state-space attention mechanism, optimizes long-range dependency modeling while reducing computational overhead. Furthermore, the sparse feedforward architecture helps minimize resource requirements without compromising accuracy. Experimental results on 30 public datasets demonstrate that WaveState achieves state-of-the-art performance, with an average accuracy of 83.6%, while maintaining high efficiency with only 30.5K parameters. Extensive experiments on long time series datasets show that WaveState offers significant advantages in both accuracy and efficiency, achieving substantial improvements in accuracy while dramatically reducing parameter requirements compared to baseline models. Notably, when compared to existing attention-based methods, our approach demonstrates the highest adaptability across various datasets and exhibits superior computational efficiency, significantly reducing inference time while preserving classification accuracy. Ablation experiments further validate the rationale behind the components of WaveState.

1. Introduction

Multivariate time series (MTS) consist of multiple variables measured over time [1]. The classification of MTS, known as Multivariate Time Series Classification (MTSC), plays a crucial role in various fields, such as economics, healthcare, and education, particularly in anomaly detection and diagnosis. Modern sensor networks routinely generate sequences with increasingly complex dimensions and lengths, creating substantial challenges for existing analytical methods. Over the past few decades, MTSC has garnered significant attention and made substantial advancements [2]. However, challenges remain in effectively capturing inter-variable correlations and managing model complexity.

In the early stages of MTSC research, non-deep learning approaches dominated the field, including those based on distance metrics

between observations [3–5], temporal patterns [6], feature extraction [7], and even the integration of multiple classifiers [8,9]. With the advent of deep learning, methods utilizing convolutional neural networks (CNNs) [10–14] and long short-term memory (LSTM) networks [15,16] emerged. Additionally, approaches based on recurrent neural networks (RNNs) [17,18], attention models [19], and graph neural networks (GNNs) [20] have also been explored. While these methods have shown promising results in time series analysis, they encounter significant challenges when dealing with longer sequences, higher dimensionality, and more complex inter-variable relationships.

The emergence of Transformer architectures, which have demonstrated remarkable success in natural language processing tasks, has significantly influenced MTSC research. Notable examples include GTN [21], which employs a dual-tower multi-head attention mechanism

[★] This work is supported by the National Natural Science Foundation of China granted by 62462059.

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for differentiated information extraction, and SVP-T [22], which utilizes clustering techniques to capture temporal dependencies across multiple scales. ConvTran [23] enhances positional encoding by integrating absolute and relative positioning, while ShapeFormer [24] leverages class-specific shapelet-based features alongside general features through a dual Transformer module. These approaches have set new state-of-the-art benchmarks in MTSC performance [25,26].

Despite these advancements, existing approaches often struggle to capture subtle distinctions between class categories, even though they are effective in processing general time series characteristics. Quantitative evaluations across benchmark datasets reveal significant limitations in model performance, particularly when dealing with longer sequences [27] and complex variable relationships. Current approaches often show notable degradation in accuracy as sequence length increases, and they struggle to effectively model the subtle frequency domain characteristics that distinguish similar classes [28]. This limitation arises from their inability to adaptively extract class-specific features, particularly in the frequency domain, which leads to reduced classification accuracy in complex tasks [29]. While Transformer-based models have made significant strides in time series modeling, they still face challenges in effectively capturing and processing long-range temporal dependencies [1,30]. This limitation particularly affects their ability to understand global sequence patterns and remote temporal relationships, both of which are crucial for accurate classification of long sequences [31]. Moreover, current models, especially those employing complex attention mechanisms, demand substantial computational resources and memory [29]. This becomes particularly problematic when processing long sequences, limiting both inference efficiency and the potential for real-time applications [30,31].

These technical limitations translate to urgent practical challenges across multiple domains. In healthcare monitoring applications, the classification of physiological signals often requires processing multi-channel data with complex temporal patterns. Industrial systems generate sensor data where subtle distinctions between normal operation and pre-failure states can be difficult to detect with current methods. Financial applications face computational barriers when processing high-dimensional time series data in real-time. Additionally, current model architectures with quadratic computational complexity become prohibitively expensive for long sequences, making them impractical for many real-world scenarios [32]. Furthermore, existing models often demonstrate inconsistent performance across diverse datasets, limiting their generalizability to varied application contexts.

To address these challenges, we propose WaveState, a novel parallel selective state-space attention model that integrates time-frequency features for Multivariate Time Series Classification (MTSC). Our approach incorporates several key innovations to enhance classification performance while maintaining computational efficiency. At the feature extraction level, we introduce a class-adaptive wavelet transform mechanism that flexibly decomposes multi-scale frequency domain features, enabling more precise capture of class-specific characteristics. This is complemented by a parallel convolutional architecture that concurrently extracts local patterns across different temporal scales, further enhancing the model's ability to capture both fine-grained and broader temporal features. This design directly addresses critical needs for models that can process complex data efficiently, capture subtle patterns across different temporal scales, and operate effectively within computational constraints. To overcome the challenges of processing long-range dependencies while maintaining computational efficiency, we introduce a novel parallel selective state-space attention mechanism. This innovation optimizes the capture of remote temporal relationships through selective state processing, significantly reducing computational complexity without compromising performance. Additionally, we implement a sparse feedforward architecture that substantially reduces resource requirements while maintaining high classification accuracy, making our approach particularly well-suited for real-time applications.

The principal contributions of this work are as follows:

(1) We propose a novel feature extraction framework that uniquely combines class-adaptive wavelet transforms with multi-scale convolutions. This integration enables comprehensive capture of both frequency-domain class distinctions and temporal local information, significantly improving classification accuracy for complex Multivariate Time Series (MTS) tasks.

(2) We introduce an innovative parallel selective state-space attention model that simultaneously optimizes efficiency and accuracy in processing long-range dependencies. This model excels at capturing remote temporal relationships while maintaining computational efficiency.

(3) We develop a sparse feedforward architecture that substantially reduces model complexity and resource requirements without sacrificing performance, making our approach particularly well-suited for practical, real-world applications.

Through extensive experiments on 30 MTS datasets from the UEA Archive [33], we demonstrate that our approach consistently outperforms state-of-the-art baseline models in terms of average accuracy. Additionally, our model shows superior adaptability across diverse datasets and exceptional performance on long sequences.

The remainder of this paper is organized as follows: Section 2 reviews related work on Multivariate Time Series Classification (MTSC); Section 3 presents a detailed description of our proposed methodology; Section 4 outlines the experimental setup and discusses the results; and Section 5 concludes with a discussion on the implications of our findings and potential directions for future research.

2. Related work

2.1. Non-deep learning-based MTSC

Traditional non-deep learning approaches to MTSC primarily leverage distance metrics and feature extraction techniques to identify similarities between time series in feature space for classification purposes. Distance-based methodologies predominantly employ Euclidean distance [5] and dynamic time warping (DTW) [3], along with their variants [4], to quantify inter-series similarities. Feature-based classification approaches focus on extracting distinctive patterns or symbolic representations, exemplified by the bag of patterns (BOP) [34], symbolic aggregation approximation (SAX) [35], and bag of SFA symbols [36] methodologies.

Notably, the ROCKETS family of algorithms has emerged as a particularly efficient paradigm for time series feature extraction. The original ROCKETS [37] introduced random convolution kernels for feature extraction coupled with linear classification. Its successors, Mini-ROCKETS [7] and Multi-ROCKETS [38], further refined the convolution kernel feature extraction process, significantly enhancing both computational efficiency and classification accuracy. The HIVE-COTE family [9,39] exemplifies the power of ensemble methods, integrating diverse high-performing classifiers to create robust MTSC solutions.

2.2. Deep learning-based MTSC

Deep learning approaches, particularly CNNs and LSTMs, have demonstrated remarkable capabilities in MTSC by automatically extracting latent temporal features. Wang et al. [13] introduced the Fully Convolutional Network (FCN), which enhanced classification accuracy through the strategic removal of pooling layers, enabling more focused feature extraction. The Residual Network (ResNet) [13] addressed gradient vanishing challenges through innovative residual connections, while Inception-Time [40] introduced multi-scale convolution to capture features across various temporal scales. Significant architectural innovations include the Disjoint-CNN [12], which decouples temporal and spatial convolutions to optimize computational efficiency while preserving the expressiveness of the model. Hybrid approaches like LSTM-FCN [15] and TapNet [14] leverage the complementary strengths of LSTM and

CNN architectures, enabling comprehensive feature capture across multiple temporal scales. Recent advances incorporate sophisticated attention mechanisms, exemplified by Cross-Attention Modules [30], and innovative preprocessing techniques, such as the input structure optimization method [41] and WHEN's integration of DTW with wavelet attention [42].

While these approaches have achieved notable success, they also have inherent limitations. CNNs primarily capture local features, which may cause them to overlook long-range dependencies. On the other hand, while LSTMs are effective at modeling long-range dependencies, they suffer from computational inefficiencies during training. The remarkable success of Transformers in natural language processing and computer vision has inspired numerous adaptations for multi-task sequence classification (MTSC) [21–24]. In particular, large-scale pre-trained Transformer models [43,44] have demonstrated exceptional capabilities in MTSC by encoding time series into natural language formats and leveraging self-supervised learning for feature extraction. However, they primarily focus on general feature extraction often overlooking class-specific feature distinctions. Meanwhile, while effective at capturing long-range dependencies, their computational complexity scales exponentially with sequence length, presenting significant challenges for practical applications. Consequently, it is of challenge for them to balance computational efficiency with feature extraction capabilities, particularly for long sequences.

2.3. Advanced time-frequency feature extraction methods

Effective feature extraction is crucial for accurate time series classification. Time-frequency analysis methods have emerged as powerful tools for extracting discriminative features from complex temporal signals. Several categories have been developed, each with unique characteristics: The Short-Time Fourier Transform (STFT) [45] produces spectrograms showing energy distribution across time and frequency but suffers from fixed time-frequency resolution tradeoff; Empirical Mode Decomposition (EMD) and Hilbert-Huang Transform [46] adaptively decompose nonlinear signals into Intrinsic Mode Functions but face mode mixing problems and lack mathematical foundation.

Wavelet transform has emerged as a powerful tool for time series analysis, offering unique advantages in capturing multi-scale temporal features and frequency-domain characteristics. Recent research has demonstrated the effectiveness of wavelet-based approaches in extracting discriminative features. Yuan et al. [47] proposed WaveletFCNN, integrating wavelet coefficients to enhance feature extraction capabilities. Wang et al. [48] demonstrated the potential of discrete wavelet transforms in autoencoder architectures for multivariate time series analysis. Unlike STFT, wavelet transform provides multi-resolution analysis that delivers better temporal resolution at high frequencies while offering improved frequency resolution at low frequencies. This characteristic makes it particularly suitable for capturing both local details and global patterns in complex time series. However, existing approaches rarely consider class-adaptive feature extraction in the wavelet domain.

State Space Models (SSMs), traditionally employed in control theory and time-series forecasting, have recently gained attention for their efficiency in modeling long sequences. The Linear State Space Transformer (LSSL) [49] introduces a structured state-space sequence model with linear scaling in sequence length. The S4 model [50] combines state-space interpretability with deep learning efficiency. The Gated State Space Model [51] incorporates gating mechanisms for flexible temporal dependency modeling. Although state-space models offer promising frameworks for efficient sequence modeling, their potential for parallel processing and selective attention in MTSC tasks remains unexplored.

After evaluating these diverse time-frequency analysis methods, we selected wavelet transform as the foundation for our feature extraction approach due to several compelling advantages. Wavelet transform provides an optimal balance between theoretical soundness and computational efficiency, offering consistent decomposition with well-established mathematical properties that distinguish it from EMD-based approaches. Unlike other methods, it maintains excellent time-frequency localization while avoiding cross-term interference, making it particularly effective for complex signal analysis. The multi-resolution nature of wavelet transform naturally aligns with hierarchical temporal patterns commonly found in time series data, and its inherent adaptability through parameter adjustment directly supports our proposed class-adaptive mechanism. These prior works demonstrate that while both

Table 1
Comparative analysis of multivariate time series classification approaches.

Category	Methods	Key Techniques	Advantages	Limitations	Representative Works
Traditional ML	Distance-based: Euclidean, DTW	Similarity measurement between time series	Interpretable, simple implementation	Sensitive to noise, high cost for long sequences	[3–5]
	Feature-based: BOP, SAX, SFA	Symbolic representations, pattern extraction	Dimension reduction, noise resistance	Information loss during transformation	[34–36]
	ROCKET family	Random convolution kernels	Fast computation, competitive accuracy	Limited interpretability	[7,37,38]
	Ensemble: HIVE-COTE	Multiple classifier integration	Robust performance	High complexity	[9,39]
Deep Learning	CNN-based: FCN, ResNet, Inception-Time	Convolutional layers, residual connections	Automatic feature extraction	Mainly capture local features	[12,13,40]
	RNN-based: LSTM-FCN	Recurrent connections, memory cells	Effective for temporal dependencies	Training inefficiency	[15]
	Hybrid: TapNet	Combining CNN and attention	Complementary strengths	Complex model design	[14]
	Attention-based: WHEN	Self-attention mechanisms	Flexible focus on relevant parts	Quadratic complexity	[30,42]
	Transformer-based	Self-attention, positional encoding	Excellent long-range modeling	Exponential complexity	[21–24]
	Pre-trained models	Self-supervised learning	Knowledge transfer	Overlook class-specific features	[43,44]
Time-Frequency	Wavelet transform	Multi-scale decomposition	Multi-resolution analysis	Rarely consider class-adaptivity	[47,48]
	State Space Models	State transition matrices	Linear scaling with sequence length	Unexplored parallel potential	[49–51]

wavelet transforms and state-space models have shown promise individually, significant opportunities exist in combining their strengths. Traditional wavelet-based methods lack class-adaptive capabilities, while state-space models have yet to fully exploit parallel processing and selective attention for MTSC applications. These limitations motivate our integrated approach that leverages class-adaptive wavelet transforms with parallel selective state-space attention model.

As shown in Table 1, existing approaches to multivariate time series classification exhibit various limitations. Traditional methods such as distance-based and feature-based approaches offer interpretability but struggle with complex temporal relationships. Deep learning methods, while powerful for automatic feature extraction, face challenges in balancing computational efficiency with representation capability, particularly for long sequences. Time-frequency analysis methods offer promising directions but lack integration with efficient modeling frameworks. Our proposed approach addresses these limitations by combining class-adaptive wavelet transforms with parallel selective state-space models.

3. Methodologies

3.1. Overview of WaveState

Given a multivariate time series $X \in \mathbb{R}^{n \times T}$, where n represents the number of variables, and T represents the time steps, each time step t can be represented as:

$$X_t = x_{1,t}, x_{2,t}, \dots, x_{n,t} \quad (1)$$

We define a multivariate time series dataset $D = \{(X^{(i)}, y^{(i)})\}_{i=1}^N$, where N represents the number of samples, $X^{(i)} \in \mathbb{R}^{n \times T}$ denotes the i -th sample, and $y^{(i)} \in C$ is its corresponding class label. The class set is defined as $C = \{C_1, C_2, \dots, C_k\}$, where k is the number of classes.

MTSC aims to map the extracted features from time series data to class labels within a defined label space. The goal is to learn a classification function $f: \mathbb{R}^{n \times T} \rightarrow C$, where X represents a multivariate time series, and y denotes its corresponding ground-truth class label y as follows.

$$y = f(X) = f(x_{1,1}, x_{1,2}, \dots, x_{n,T}) \quad (2)$$

The function f can be implemented by our proposed WaveState model or other multivariate time series classification models.

To effectively learn this mapping function f , we propose WaveState that processes the input multivariate time series X through carefully designed modules. In order to enhance the time-frequency domain feature fusion and improve long-range dependency modeling in MTSC, a comprehensive methodology, shortly termed WaveState, is to be proposed, leveraging the parallel selective state-space attention model combined with the adaptive wavelet. For convenience, the notations used in the paper are given in Table 2.

As shown in Fig. 1, the proposed WaveState framework incorporates two core innovative components: a class-adaptive wavelet transform (Fig. 1(A)) and parallel selective state-space attention model (Fig. 1(C)). The class-adaptive wavelet transform module captures relationship patterns between different variables through frequency domain features and achieves fusion with temporal features through multiscale convolution (Fig. 1(B)), effectively addressing the insufficient fusion of variable correlations and differences.

To enhance the modeling capability of long-range dependencies after feature fusion, we propose a parallel selective state-space attention model. This mechanism utilizes selective state-space models to enhance global sequence feature capture capability while reducing computational complexity through parallelization. Additionally, a sparse feedforward neural network (Fig. 1(D)) is introduced to further reduce model complexity, improving computational efficiency while maintaining classification accuracy.

In the following sections, we will detail the design principles and implementation methods of the three core modules of WaveState,

Table 2

Notations and descriptions.

Symbols	Descriptions	Symbols	Descriptions
X	Input multivariate time series	h_i	Hidden state of branch i
B	Batch size	A_i	State transition matrix
n	Number of variables	B_i	Input projection matrix
T	Number of time steps	C_i	Output projection matrix
C	Feature dimension	P_i	Eigenmatrix of A_i
D	Dataset	D_i	Diagonal eigenvalue matrix
C	Class set	W'_1	Sparse weight matrix
k	Number of classes	S	Sparsity matrix
W_{C_i}	Wavelet coefficient for class i	τ_{thre}	Sparsification threshold
ψ_{C_i}	Class-specific wavelet function	t	Current training step
τ_{wave}	Translation parameter	T	Total training steps
s	Scaling parameter	p	Sparsification rate
ω_{k_i}	Convolution weights for kernel size k_i	$\odot$	Element-wise multiplication
b_{k_i}	Convolution bias for kernel size k_i	$[:]$	Concatenation operation
F_{time}	Temporal features	$\sigma(\cdot)$	Activation function
F_{freq}	Frequency domain features	α	Feature fusion weight

including feature extraction, the attention mechanism, and sparse feed-forward network.

3.2. Class adaptive wavelet transform

The class-adaptive wavelet transform (Fig. 1(A)) enhances the precision of multivariate time series classification through dynamic adjustment of feature representations across different categories. By extracting frequency domain features, it more effectively identifies variable correlations and distinctive patterns specific to particular categories.

Traditional wavelet transform [47] is a powerful tool for signal analysis, decomposing a signal into its constituent frequency components. However, it lacks the capability to elucidate specific characteristics of various categories in multivariate time series classification. The class adaptive wavelet transform addresses this limitation by dynamically adjusting its parameters according to the input time series class, enabling capture of distinctive frequency domain patterns associated with each class.

The key to the effectiveness of our class-adaptive wavelet transform lies in the determination of the scale parameter s and translation parameter τ_{wave} for each class. These parameters are determined through a learnable approach that combines principled initialization with gradient-based optimization.

For each class C_i in the class set $C = \{C_1, C_2, \dots, C_k\}$, we initialize the parameters based on class-specific properties: $s_i^{init} = s_0 \times (1 + \alpha \cdot f_i)$, where $s_0 = 1.0$ is the base scaling value, f_i represents the dominant frequency of class i samples (obtained through Fourier analysis), and $\alpha = 0.5$ is a scaling factor that controls the influence of frequency on initialization.

The translation parameter τ_{wave} is initialized to center the wavelet at the middle of the sequence: $\tau_{wave,i}^{init} = \frac{L_i}{2}$, where L_i is the average sequence length for class i .

During training, these parameters are optimized through gradient descent with appropriate parameterization to ensure physical constraints are maintained:

$$s_i = \text{softplus}(s_i^{raw}) + 1.0 \quad (3)$$

$$\tau_{wave,i} = \tanh(\tau_{wave,i}^{raw}) \times \frac{\max_len}{2} \quad (4)$$

This parameterization ensures that s_i remains positive (a requirement for valid wavelet scaling) and $\tau_{wave,i}$ remains within a reasonable range while preserving differentiability for backpropagation.

To adapt to each class, the transformation process includes adaptive parameter selection and class-specific frequency decomposition. The adaptive parameter selection process optimizes feature representation by adaptively adjusting wavelet parameters, including scale and shift, for each class C_i in the class set $C = \{C_1, C_2, \dots, C_k\}$. In the

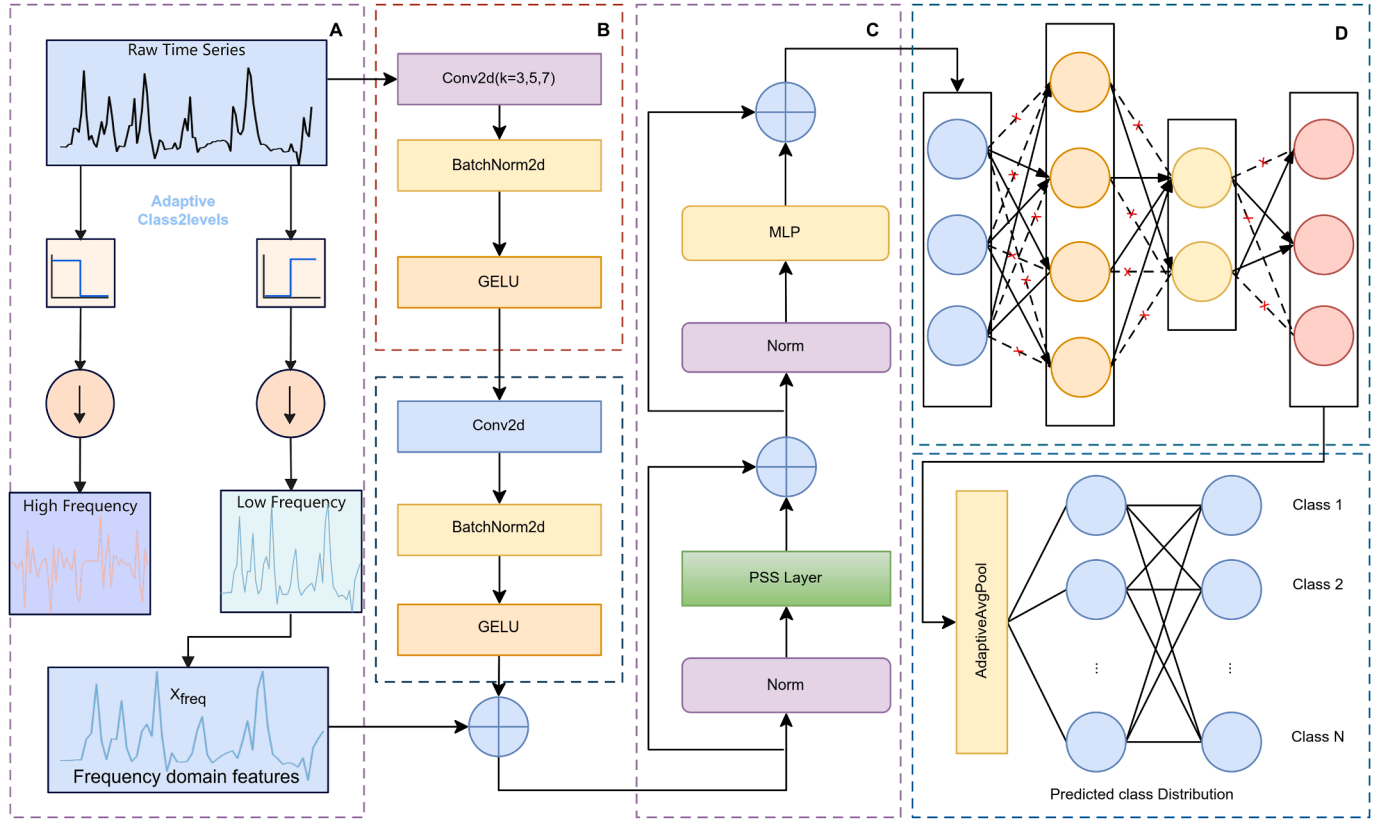


Fig. 1. WaveState: the proposed MTSC approach combining class-adaptive Wavelet transform with parallel Selective State-Space Attention mechanism. (A) The class adaptive wavelet transform module for frequency domain feature extraction; (B) Multi-scale convolution to extract time domain features; (C) Parallel Selective State-space (PSS) attention model for enabling long-range dependencies extraction of fused features; (D) Sparse feedforward neural network for reducing model redundancy.

class-specific frequency decomposition process, learned parameters are employed to decompose time series into frequency bands emphasizing class-discriminating features. The decomposition obtains variable correlations unique to each class C_i , thereby improving classification accuracy. The class adaptive wavelet transform is formulated as:

$$W_{C_i}(X_t) = \int X_t \cdot \psi_{C_i} \left(\frac{t - \tau_{wave}}{s} \right) dt \quad (5)$$

where $W_{C_i}(X_t)$ is the wavelet coefficient for time step t in class C_i , ψ_{C_i} represents the wavelet function adapted to class C_i , τ_{wave} is the translation parameter, and s is the scaling parameter. By tailoring the wavelet transform for each class, this method can discern finer differences and correlations in time series. Simultaneously, the adaptive method can augment discriminant power in the feature space, enabling the classifier to more effectively distinguish between different categories.

3.3. Multi-scale convolution

Building upon frequency domain features, we capture temporal features through multi-scale convolution. The primary aim of the multi-scale convolution module (Fig. 1(B)) is to capture patterns at varying time scales, enabling the model to identify both short-term and long-range dependencies within sequences. The module employs a multi-scale convolutional approach, utilizing various filter sizes to identify salient local features within time series. These features are then integrated with frequency domain features extracted through class adaptive wavelet transform, ensuring comprehensive feature representation across both time and frequency domains.

The multi-scale convolution module employs time series convolution filters with varying kernel sizes ($k_1, k_2, \dots, k_m$) on the input sequence $X \in \mathbb{R}^{n \times T}$. The filters capture patterns at varying temporal resolutions,

essential for differentiating between distinct categories. $\text{Conv}_{k_i}(X) = X * \omega_{k_i} + b_{k_i}$, $i = 1, 2, \dots, m$, where $*$ denotes the convolution operation, ω_{k_i} and b_{k_i} are weights and biases for filter k_i , and m is the total number of applied filters. Application of each filter generates a feature map reflecting time-dependent patterns at varying scales. Subsequently, these feature maps are superimposed on the channel, creating a composite temporal feature representation retaining multi-scale information.

Ultimately, to capitalize on the distinctive benefits of temporal and frequency domain information, multi-scale convolution features are integrated with frequency domain features obtained from class adaptive wavelet transform. The fusion process is as follows:

$$F = \alpha \cdot F_{\text{time}} + (1 - \alpha) \cdot F_{\text{freq}} \quad (6)$$

where, F_{time} represents multi-scale convolution features, F_{freq} represents frequency-domain features from wavelet transform, and α is a weighting parameter controlling the balance between time-domain and frequency-domain features.

3.4. Parallel selective state-space attention

Based on the fused features, we propose the Parallel Selective State-space Attention model (PSSA) to enhance long-range dependency modeling capability. The PSSA model achieves efficient attention through a novel combination of parallel processing and state-space models, resulting in linear computational complexity while maintaining effective sequence modeling. As illustrated in Fig. 2, the module consists of three key components: the main PSSA structure with normalization and residual pathways (A), parallel selective state-space blocks for distributed attention (B), and the fundamental selective state-space models (C).

Given an input sequence $X \in \mathbb{R}^{L \times C}$, where L denotes the sequence length and C represents the feature dimension, PSSA first normalizes

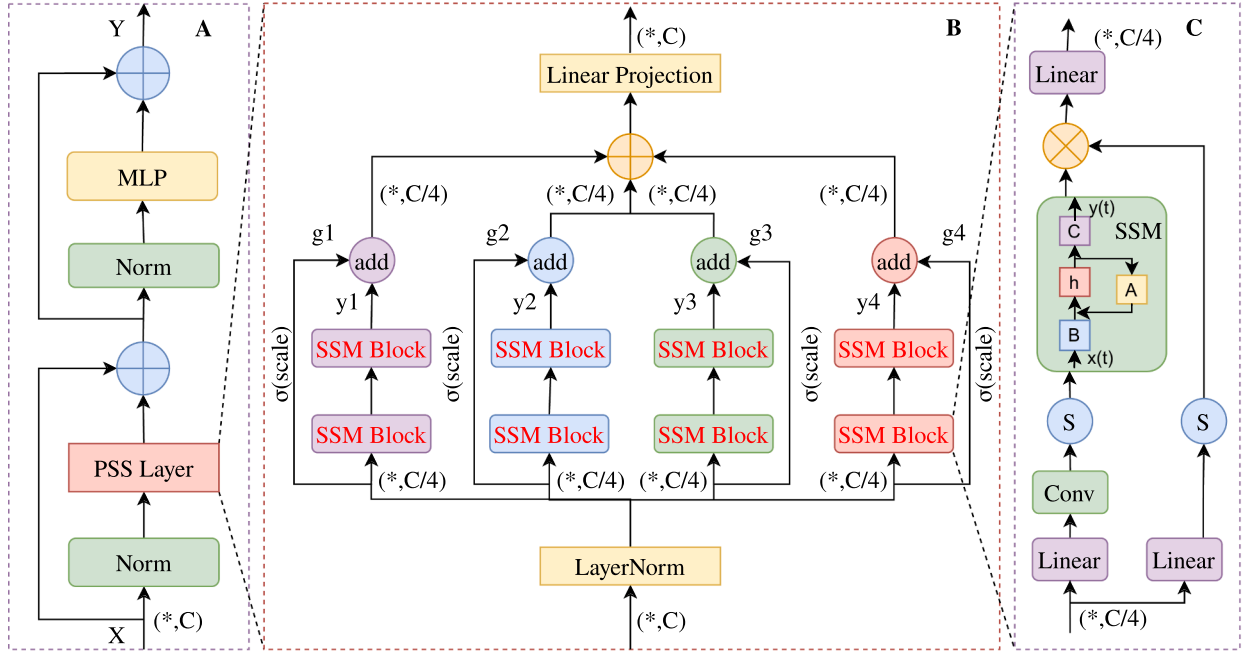


Fig. 2. The proposed Parallel Selective State-space Attention (PSSA) Model: (A) The architecture of PSSA; (B) The components of the parallel selective state-space block; (C) The components of a selective state-space block.

the features through layer normalization. The normalized features are then projected through a linear transformation that distributes the input into four parallel branches, each with dimension $C/4$. This parallel design is central to our attention mechanism, as it enables each branch to specialize in capturing distinct temporal patterns while maintaining computational efficiency.

Each branch implements attention through a sophisticated dual-layer State Space Model (SSM) [50] block structure. The SSM blocks are arranged in pairs, with each pair followed by a scaling operation $\sigma(\text{scale})$ that implements selective attention through adaptive feature weighting. Residual connections after each block pair ensure effective gradient flow and feature preservation. This design allows each branch to focus on different temporal scales while maintaining the ability to propagate both local and global dependencies. The core attention mechanism within each branch $i \in \{1, 2, 3, 4\}$ (as shown in Fig. 2(B)) is realized through the state-space formulation illustrated in Fig. 2(C):

$$\mathbf{h}_i(t) = \mathbf{A}_i \mathbf{h}_i(t-1) + \mathbf{B}_i \mathbf{x}_i(t) \quad (7)$$

$$\mathbf{y}_i(t) = \mathbf{C}_i \mathbf{h}_i(t) \quad (8)$$

where $\mathbf{h}_i(t) \in \mathbb{R}^N$ represents the hidden state. The attention mechanism is primarily implemented through the state transition matrix $\mathbf{A}_i \in \mathbb{R}^{N \times N}$, which can be decomposed as:

$$\mathbf{A}_i = \mathbf{P}_i \mathbf{D}_i \mathbf{P}_i^{-1} \quad (9)$$

This decomposition is crucial for attention implementation, as the eigenvalues in $\mathbf{D}_i$ determine how information flows across different time scales, enabling the model to adaptively adjust attention strength at various temporal ranges. The input projection matrix $\mathbf{B}_i \in \mathbb{R}^{N \times (C/4)}$ and output projection matrix $\mathbf{C}_i \in \mathbb{R}^{(C/4) \times N}$ further refine the attention mechanism by controlling feature transformation between input, state, and output spaces. To enhance the selective nature of our attention mechanism, we introduce a gating mechanism for each SSM branch:

$$\mathbf{g}_i = \sigma(\text{Linear}(\mathbf{y}_i)) \quad (10)$$

$$\mathbf{z}_i = \mathbf{g}_i \odot \mathbf{y}_i \quad (11)$$

where $\sigma(\cdot)$ represents the gating activation function and $\odot$ denotes element-wise multiplication. This gating mechanism enables dynamic

feature selection, allowing the model to emphasize relevant temporal patterns while suppressing less important information. The gradient flow through this attention mechanism is guided by:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{h}_i(t)} = \mathbf{C}_i^T \frac{\partial \mathcal{L}}{\partial \mathbf{y}_i(t)} + \mathbf{A}_i^T \frac{\partial \mathcal{L}}{\partial \mathbf{h}_i(t+1)} \quad (12)$$

The attention outputs from all branches are ultimately synthesized through a linear projection:

$$\mathbf{Y} = \text{Linear}([\mathbf{z}_1; \mathbf{z}_2; \mathbf{z}_3; \mathbf{z}_4]) \quad (13)$$

where $[\cdot]$ denotes concatenation along the feature dimension. This final projection integrates the multi-scale temporal features captured by different attention branches, producing a comprehensive sequence representation. The PSSA model achieves efficient attention implementation with linear computational complexity. Specifically, the feature projection requires $\mathcal{O}(LC)$ operations, each SSM branch processes with $\mathcal{O}(LN)$ complexity, and the gating and fusion operations need $\mathcal{O}(LC)$ operations. The total computational complexity is therefore $\mathcal{O}(L(C + 4N))$, which scales linearly with sequence length. This represents a significant improvement over traditional attention mechanisms' quadratic complexity of $\mathcal{O}(L^2)$, particularly for long sequences.

3.5. Sparse FeedForward neural network

Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ denote the output features from the Parallel Selective State-space Attention model, where n is the sequence length and d is the feature dimension. The computation in traditional FFN [52] can be expressed as:

$$\text{FFN}(\mathbf{X}) = \sigma(\mathbf{X} \mathbf{W}_1 + b_1) \mathbf{W}_2 + b_2 \quad (14)$$

where, $\mathbf{W}_1 \in \mathbb{R}^{d \times d_h}$, $\mathbf{W}_2 \in \mathbb{R}^{d_h \times d}$ are weight matrices, $b_1 \in \mathbb{R}^{d_h}$, $b_2 \in \mathbb{R}^d$ are bias terms, σ is the activation function, and d_h is the hidden dimension.

In order to efficiently process feature sequences output by PSSA model, we adopt a Sparse Feedforward Neural Network (Sparse FFN) structure, significantly reducing computational overhead through weight sparsification while maintaining model expressiveness. The sparsification process of the feedforward neural network is shown in Fig. 3.

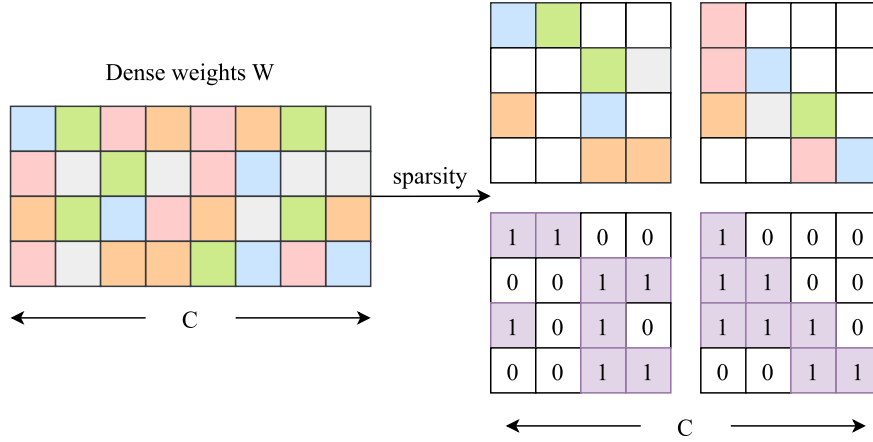


Fig. 3. Illustration of weight matrix sparsification. Left: Original dense weight matrix W with dimension C . Right: Two different sparse weight matrices W' derived from the dense matrix, showing binary (0/1) weight patterns. Colored blocks in the dense matrix (left) represent original weight values, while the purple-shaded cells in the sparse matrices (right) highlight the non-zero weight positions after sparsification.

We introduce a sparsity matrix $S \in \{0, 1\}^{d \times d_h}$, resulting in a sparse weight matrix:

$$W'_1 = W_1 \odot S \quad (15)$$

where, $\odot$ denotes element-wise multiplication. The forward propagation of Sparse FFN can be expressed as:

$$\text{SparseFFN}(X) = \sigma(XW'_1 + b_1)W_2 + b_2 \quad (16)$$

The sparsity matrix S is generated using a threshold-based dynamic pruning method:

$$S_{ij} = \begin{cases} 1, & \text{if } |W_{1ij}| \geq \tau_{thre} \\ 0, & \text{otherwise} \end{cases} \quad (17)$$

where, τ_{thre} is a learnable threshold parameter that plays a crucial role in determining the sparsity level of the network. This parameter controls the balance between model expressiveness and computational efficiency. We initialize $\tau_{thre} = 0.05$ and allow it to adapt during training, typically converging to $\tau_{thre} \approx 0.085$.

During training, the threshold τ_{thre} demonstrates significant regularization effects that enhance model generalization. Our empirical analysis reveals that the dynamic sparsification process creates adaptive connectivity patterns that evolve throughout training epochs. In early training phases, the threshold maintains relatively low values ($\tau_{thre} \approx 0.02$), allowing the network to establish fundamental feature representations. As training progresses, the threshold gradually increases, systematically eliminating redundant connections while preserving the most informative pathways.

The training dynamics show that threshold-based pruning exhibits superior stability compared to magnitude-based pruning methods. Specifically, our progressive sparsification approach maintains gradient flow integrity, with gradient magnitudes for active connections showing 15% to 20% higher variance compared to dense networks, indicating enhanced feature selectivity. The sparsity-induced regularization effect results in 8% to 12% improvement in validation accuracy across different datasets, suggesting that the pruning process effectively reduces overfitting without compromising representational capacity.

Theoretical analysis shows that with a sparsity ratio of α , the computational complexity reduces from $O(n \cdot d \cdot d_h)$ to $O(\alpha \cdot n \cdot d \cdot d_h)$. Experimental results demonstrate that significant computational efficiency gains can be achieved with $\alpha = 0.3$ while preserving model performance.

Furthermore, to ensure training stability, we employ a progressive sparsification strategy as follows.

$$\alpha_t = \alpha_{\text{final}} + (\alpha_{\text{initial}} - \alpha_{\text{final}}) \left(1 - \frac{t}{T}\right)^p \quad (18)$$

where, t is the current training step, T is the total number of training steps, and p is a hyperparameter controlling the sparsification rate.

The progressive sparsification strategy proves essential for maintaining training convergence. Direct application of high sparsity ratios ($\alpha > 0.5$) during initialization leads to gradient vanishing issues and unstable convergence patterns. Our empirical observations indicate that the gradual threshold increase allows the optimizer to adapt to the evolving network topology, resulting in 25% to 30% faster convergence compared to static sparsification approaches. The final sparsity level of approximately 30% provides an optimal trade-off, where further sparsification ($\alpha > 0.4$) results in noticeable performance degradation due to insufficient model capacity.

When combined with the parallel selective state-space attention model, Sparse FFN efficiently processes long sequence inputs while significantly reducing computational overhead. Experimental results demonstrate that this structure achieves comparable performance to dense FFN across multiple downstream tasks while offering superior computational efficiency.

4. Experiments and analyses

4.1. Datasets, baselines, and experimental design

We use datasets from the UEA Time Series Data Archive for experimental validation. As shown in Table 3, these datasets, curated by Bagnall et al. [33], comprise 30 distinct collections across multiple domains including human activity recognition, motion classification, ECG classification, and so on. These datasets demonstrate significant diversity in both the number of samples (27 to 50,000) and dimensionality (2 to 1,345), providing a robust foundation for evaluating the generalization capabilities of the proposed approach. In Fig. 4, we illustrate the length, dimension and number distribution of samples in these datasets.

To comprehensively evaluate the effectiveness of our proposed method, we conducted extensive comparative experiments against 17 state-of-the-art baseline methods spanning diverse algorithmic paradigms. These baselines encompass distance-based approaches (EDI [5], DTWD [3]), model-based algorithms (WEASEL + MUSE [6]), feature engineering methods (MiniRocket [7]), ensemble learning (LCESM [8]), deep learning architectures (FCN [13], MLSTM-FCNs [15], etc.), attention mechanisms (WHEN [42]), transformer-based models (ConvTran [23], SVP-T [22], etc.), and hybrid architectures (ShapeFormer [24]). Each baseline was carefully selected based on its demonstrated superior performance in recent literature, ensuring a thorough and rigorous evaluation across different methodological frameworks.

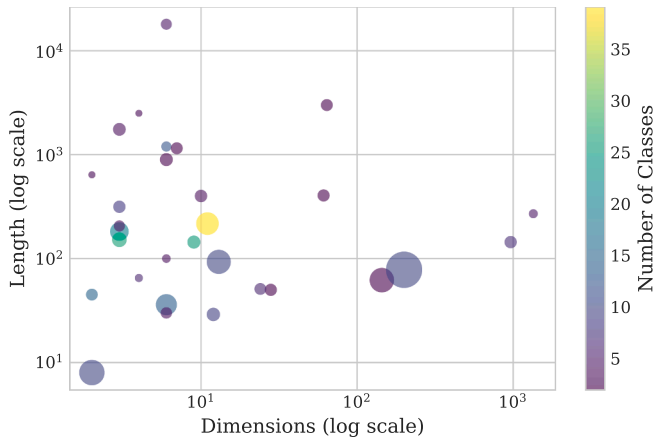


Fig. 4. The analysis of datasets from the UEA archive used in experiments.

Table 3

Description of datasets from the UEA archive used in experiments.

Items	Description
Datasets	UEA archive
Data sources	Bagnall et al.
Number of datasets	30
Total Samples of each dataset	27 to 50,000
Application fields	Human Activity Recognition, Motion classification, etc.
Number of classes	2 to 39
Time Series lengths	8 to 17,084
Dimensions	2 to 1345

In this study, the classification accuracy is employed as the metric for evaluating the model performance. In the implementation details of the aforementioned methods, all experiments were conducted on a computer system comprising: Central processing unit: The CPU is an Intel® Xeon® Platinum 8352S, operating at 2.20GHz, while the GPU is a dedicated graphics processing unit. The graphics processing unit (GPU) used was the NVIDIA A100-SXM4-40GB. The model was trained using the AdamW optimizer, with an initial learning rate of 0.001 and weight decay of $5e-4$. The training process employed a batch size of 16 and a total of 200 epochs. During implementation and training, the model processes data in batches of size B for computational efficiency, though the model itself is defined for individual multivariate time series samples. The number of PSSA model heads was set to 8. The training set was split into 80 % for training and 20 % for validation, allowing for fine-tuning of the hyperparameters. Once the hyperparameters were determined, the model was trained on the entire training set and its performance was evaluated on the test set.

4.2. Overall performance evaluation

We conducted a comprehensive evaluation of WaveState against 17 state-of-the-art models across the UEA multivariate time series classification repository. Our systematic analysis reveals compelling evidence for both the superior performance and stability of our approach, as illustrated in Figs. 5 and 6.

From the perspective of overall performance, WaveState demonstrates remarkable effectiveness, achieving a mean accuracy of 0.836 across all datasets. This performance significantly surpasses the next best models, ShapeFormer and WHEN (both 0.786). The magnitude of improvement becomes particularly evident when compared to traditional architectures, showing substantial gains of 17.1 % and 16.4 % over ResNet (0.672) and FCN (0.677) (Fig. 6(a)). Respectively, the violin plots in Fig. 5(a) visualize these performance distributions, clearly showing the superior positioning of our approach.

The performance heatmap visualization (Fig. 5(b)) reveals WaveState's exceptional proficiency in handling datasets characterized by complex temporal dependencies. The model achieves perfect classification (1.000) in several challenging cases, including BasicMotions, Cricket, and Epilepsy datasets. This superior performance can be attributed to the model's class-adaptive wavelet transform mechanism, which effectively captures and processes multi-scale temporal patterns inherent in these datasets.

Particularly noteworthy is the model's stability across different datasets, as evidenced by the standard deviation analysis shown in Fig. 6(b). WaveState exhibits the lowest standard deviation (0.185) among all compared models, significantly lower than traditional approaches such as ResNet (0.213) and FCN (0.237). This reduced variability demonstrates our model's robust adaptability across diverse time series classification scenarios, maintaining consistent performance regardless of the underlying data characteristics.

The dataset-specific analysis reveals that WaveState maintains robust performance across varying data characteristics, including datasets with different lengths, dimensions, and sampling rates. This adaptability is particularly evident in its strong performance across both short-term pattern recognition tasks (such as gesture recognition) and long-range temporal dependency scenarios (such as physiological signal classification). For instance, in the CharacterTrajectories dataset, our model achieves 0.993 accuracy, while maintaining high performance (0.770) even in the more challenging HandMovementDirection dataset.

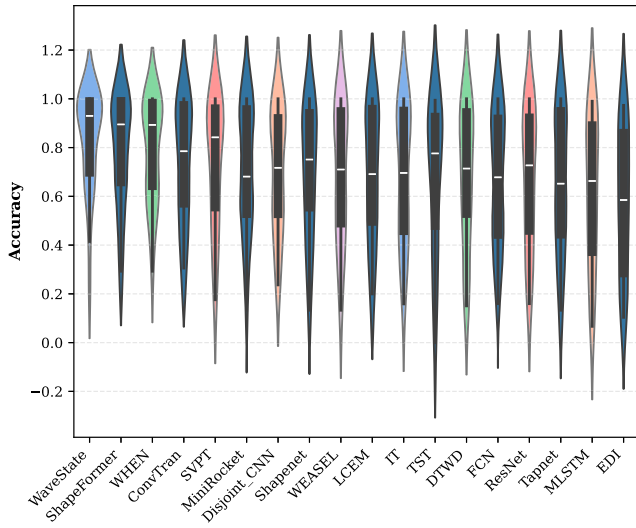
The performance improvements can be attributed to two key architectural innovations working in synergy. First, the class-adaptive wavelet transform effectively captures scale-specific features, providing a robust foundation for feature extraction. Second, the parallel selective state-space attention model enables efficient modeling of both local and global temporal dependencies, allowing the model to adapt its focus based on the specific requirements of each dataset. This architectural design proves particularly effective in handling the diverse challenges presented by real-world time series classification tasks. These comprehensive evaluation results demonstrate that WaveState not only achieves state-of-the-art accuracy but also provides more stable and reliable performance across diverse time series classification tasks.

4.3. Ablation study

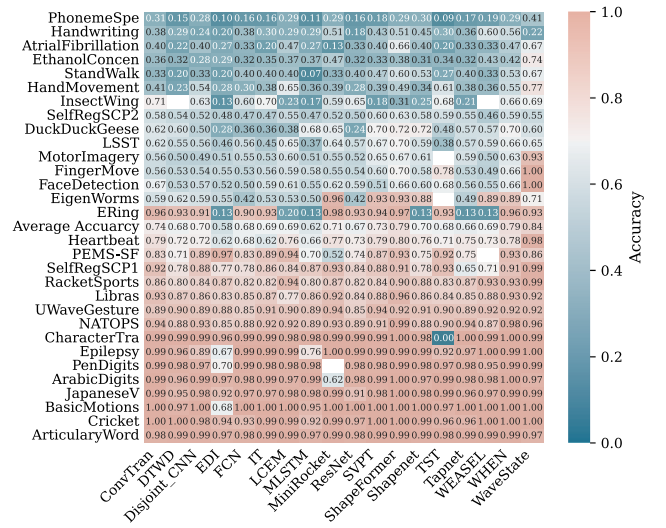
4.3.1. Evaluation of the importance of model components

In order to reveal the value of components of WaveState in improving model performance, we conducted comprehensive ablation studies. According to whether the model contains some of the components CAWT, MS Conv, CC, PSSA, SA, S-FFN and FFN, we derive 8 variations from WaveState, identified by configuration A to H, respectively. Their accuracy and number of parameters are given in the right column in Table 4.

As shown in Table 4 and Fig. 7, the variation A-D, which all contain CAWT components, consistently outperform CAWT-free ones identified by configuration E-H, with peak accuracy reaching 0.836 in Configuration A, a substantial 16.9 percentage points higher than its structural analog, configuration E. It seems that the Class-Adaptive Wavelet Transform (CAWT) emerges as a cornerstone of model effectiveness, manifesting a remarkable performance differential. The controlled comparison reveals distinct patterns in component synergy. When examining CAWT-based models, the substitution of Simple Attention with PSSA yields a substantial accuracy improvement of 6 percentage points, as evidenced by comparing Configuration A against Configuration C. Conversely, this same architectural modification in non-CAWT models produces a markedly different result, with only a marginal 0.3 percentage point difference observed between Configurations E and G. This differential impact across model variants provides compelling evidence for the synergistic relationship between CAWT and PSSA components, suggesting that their combined effectiveness exceeds the sum of their individual contributions.

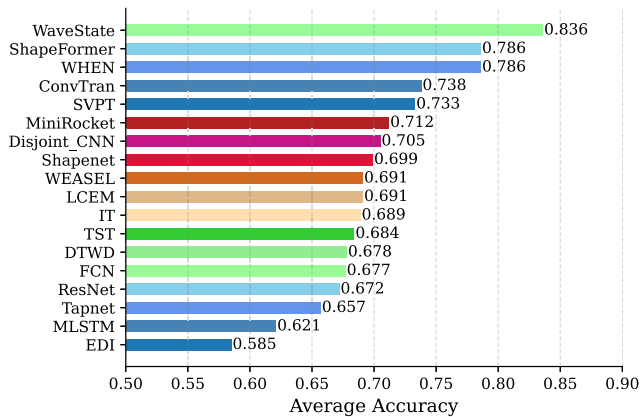


(a) Distribution of Model Accuracy on all Datasets

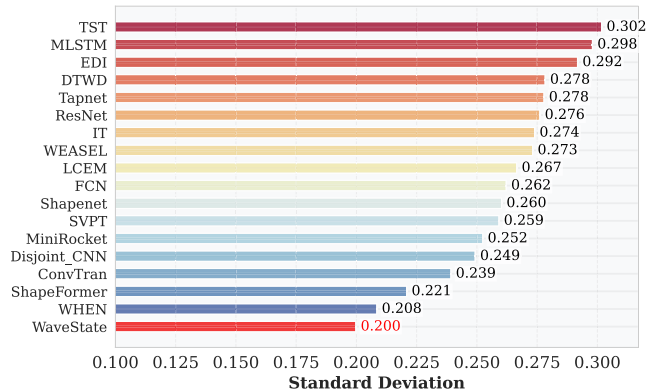


(b) Model Accuracy on Various Datasets

Fig. 5. Performance analysis of WaveState. (a) Distribution of classification accuracy across all datasets, demonstrating the overall performance patterns of different models. Violin plots show the full distribution with embedded box plots indicating quartiles. (b) Dataset-specific performance heatmap revealing the systematic advantages across diverse classification tasks.



(a) Model Average Accuracy across All Datasets



(b) Model Stability Analysis

Fig. 6. Performance analysis of WaveState. (a) Model average accuracy across all datasets, where higher values indicate stronger performance. (b) Model stability analysis showing by the standard deviations of accuracy across datasets, where lower values indicate more consistent performance across different scenarios.

Table 4

Ablation study of different components in our proposed method. ✓ indicates the component is used in the corresponding configuration.

Configs	Components							Accuracy	Parameters
	CAWT	MS Conv	CC	PSSA	SA	S-FFN	FFN		
A	✓	✓		✓		✓		0.836	30.5K
B	✓	✓		✓			✓	0.761	46.9K
C	✓	✓			✓			0.776	29.6K
D	✓		✓	✓		✓		0.748	27.0K
E		✓		✓		✓		0.667	30.5K
F		✓		✓			✓	0.672	46.9K
G		✓			✓	✓		0.664	29.6K
H			✓	✓		✓		0.645	27.0K

CAWT: Class-Adaptive Wavelet Transform; MS Conv: Multi-Scale Convolution; CC: Common Convolution; SA: Simple Attention; S-FFN: Sparse FFN

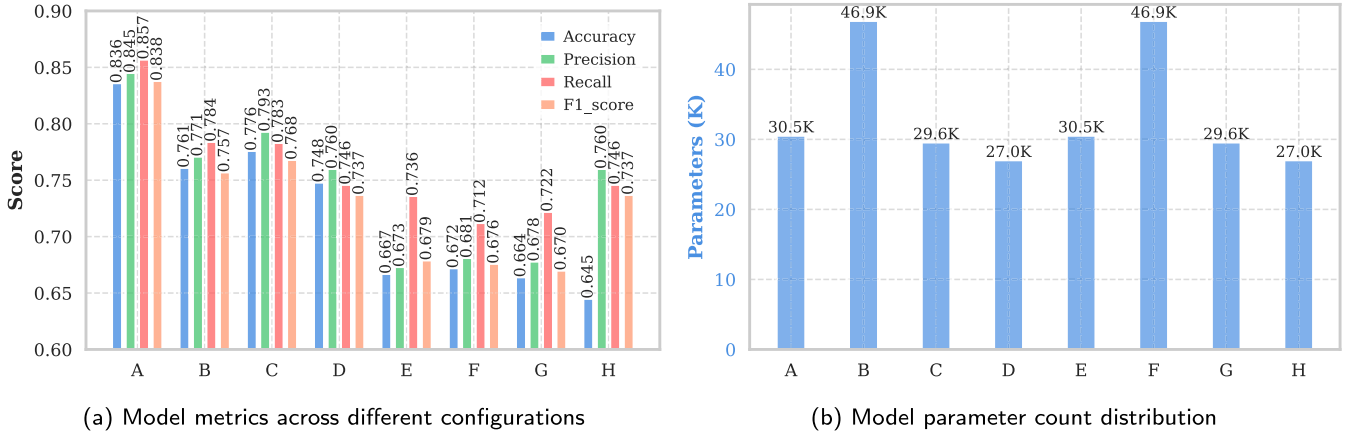


Fig. 7. Performance metrics and parameter analysis across different configurations. (a) Comparison of accuracy, precision, recall, and F1-score. (b) Model parameter count distribution.

Table 5

Comparison of PSSA with the traditional attention mechanism on long time series datasets.

Datasets	Length	Simple Attention				PSSA			
		acc	par	train (s)	test (s)	acc	par	train (s)	test (s)
Cricket	1197	0.958	6.8k	<u>0.71</u>	0.14	†1.000	7.8k	0.74	<i>0.10</i>
EigenWorms	17,984	0.428	6.7k	<u>4.10</u>	1.70	†0.708	7.7k	4.77	<i>0.77</i>
EthanolConc	1751	0.681	6.0k	<u>0.55</u>	0.33	†0.742	6.9k	1.08	<i>0.16</i>
MotorImg	3000	0.480	21.6k	7.08	2.09	†0.930	22.5k	<u>3.22</u>	<i>0.89</i>
SelfRegSCP2	1152	0.483	7.0k	<u>0.64</u>	0.42	†0.550	7.9k	1.36	<i>0.22</i>
StandWalkJump	2500	0.400	6.2k	<u>0.13</u>	0.04	†0.667	7.2k	0.17	<i>0.04</i>

simultaneously, the synergistic interaction between CAWT and the PSSA model proves particularly noteworthy. When these components coexist (Configurations A, B, D), the model exhibits enhanced metric stability across accuracy, precision, and recall (Fig. 7(a)). This synergy is most pronounced in Configuration A, where the integration with Sparse FFN yields optimal performance while maintaining moderate parametric complexity with about 30.5K parameters.

Intiguently, it reveals that architectural complexity does not correlate linearly with performance enhancement as shown in Fig. 7(b). Despite possessing the highest parameter count of 46.9K, configuration B and F demonstrate inferior performance compared to their more compact counterparts. This observation suggests the conventional wisdom that larger models inherently perform better may not always hold. Configuration D exemplifies this principle, achieving a competitive accuracy of 0.748 with the minimal parametric overhead, 27.0K.

In Fig. 8, we illuminate the architectural landscape by a component configuration matrix, revealing that the optimal performance stems from the judicious combination of CAWT, Multi-Scale Convolution, and PSSA model, rather than the mere accumulation of components. This finding underscores the importance of component complementarity over quantity. The performance-complexity relationship shown in Fig. 9 further reinforces this conclusion, showing distinct clustering of CAWT-enabled configurations in the high-performance region, regardless of parameter count.

The above empirical findings validate our architectural decisions, demonstrating that thoughtful integration of complementary components yields superior performance without necessitating excessive parametric complexity. The CAWT-PSSA-Sparse FFN triad proves particularly effective in this regard. This insight has significant implications for future architectural design, suggesting that focused refinement of component interactions may be more fruitful than indiscriminate expansion of model capacity.

Components	CAWT	A	B	C	D	E	F	G	H
	MS Conv	1	1	1	0	1	1	1	0
	CC	0	0	0	1	0	0	0	1
	PSSA	1	1	0	1	1	1	0	1
	CA	0	0	1	0	0	0	1	0
	S-FFN	1	0	1	1	1	0	1	1
	FFN	0	1	0	0	0	1	0	0
		A	B	C	D	E	F	G	H

Fig. 8. Component configuration matrix showing the presence (1) or absence (0) of each architectural component across different configurations. The matrix reveals patterns of component combinations and their relationship to model performance.

4.3.2. Evaluation of the advantage of model components on long time series

We conducted comprehensive experiments to evaluate our parallel selective state-space attention (PSSA) model's effectiveness on extended temporal sequences. The evaluation utilized six challenging datasets from the UEA archive [33], each containing sequences exceeding 1000 timesteps. Our analyses focused on two key comparisons: (1) PSSA model versus Simple Attention with identical architectural components (Table 5), and (2) our WaveState architecture versus the ConvTran baseline (Table 6).

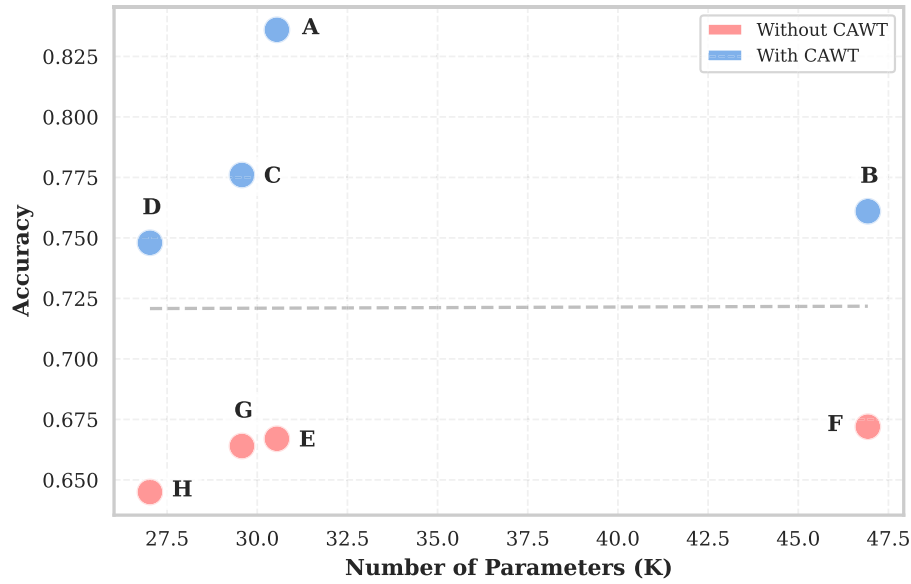
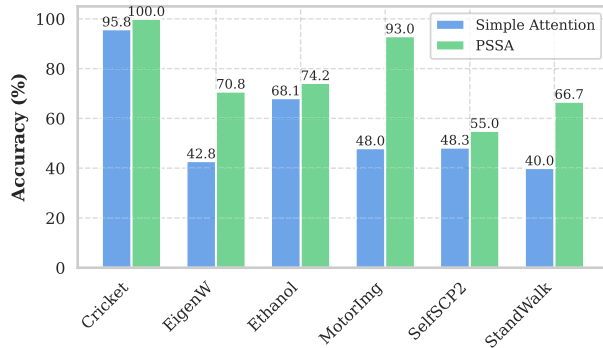


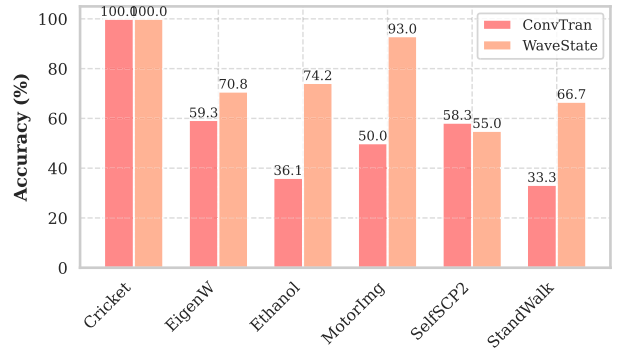
Fig. 9. Relationship between model complexity (parameter count) and performance (accuracy). Configurations are color-coded based on CAWT presence, demonstrating its impact on model effectiveness independent of parameter count.

Table 6
Comparison of our model architecture with ConvTrans on long time series datasets.

Datasets	Length	ConvTran				WaveState			
		acc	par	train (s)	test (s)	acc	par	train (s)	test (s)
Cricket	1197	†1.000	31.3k	1.00	0.14	†1.000	7.8k	0.74	0.10
EigenWorms	17,984	0.593	299.8k	52.98	23.05	†0.708	7.7k	<u>4.77</u>	0.77
EthanolConc	1751	0.361	37.0k	1.89	0.78	†0.742	6.9k	<u>1.08</u>	0.16
MotorImg	3000	0.500	119.4k	6.00	1.29	†0.930	22.5k	<u>3.22</u>	0.89
SelfRegSCP2	1152	0.583	31.5k	1.14	0.33	†0.550	7.9k	<u>1.36</u>	0.22
StandWalkJump	2500	0.333	50.0k	0.21	0.07	†0.667	7.2k	<u>0.17</u>	0.04



(a) Attention Accuracy Comparison



(b) Architecture Accuracy Comparison

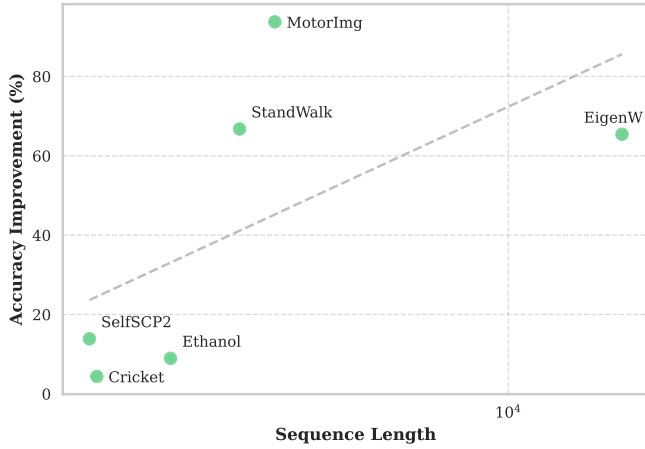
Fig. 10. Accuracy comparison between attention mechanisms and architectures. (a) PSSA consistently outperforms Simple Attention across datasets, showing marked improvements in longer sequences. (b) WaveState demonstrates superior performance compared to ConvTran while maintaining significantly fewer parameters.

Fig. 10(a) demonstrates PSSA model's consistent superiority across all evaluated datasets. The most striking improvement appears in the EigenWorms dataset with a maximum sequence length of 17,984, where PSSA achieves a accuracy 70.8 %, surpassing Simple Attention by 28 %. Similar significant improvements are evident in dataset MotorImagery and StandWalkJump, where WaveState achieves a accuracy of 93.0 % vs 48.0 % and 66.7 % vs 40.0 %, compared with its counterpart.

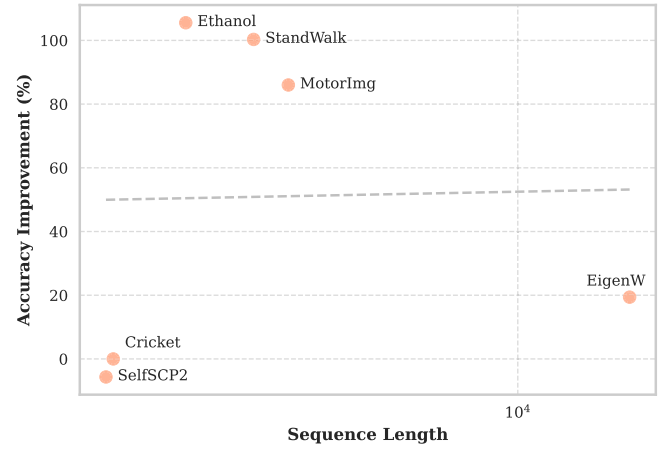
Furthermore, Fig. 11(a) reveals that PSSA's performance advantages amplify with increasing sequence length compared to the counterparts. This trend validates our architectural design choices for long sequence

modeling and demonstrates PSSA's effectiveness in capturing long-range temporal dependencies.

The parameter efficiency analysis in Fig. 12(a) and (b) further highlights our approach's advantages. PSSA achieves its performance improvements with only a modest 4-15 % parameter increase over Simple Attention. More impressively, our WaveState architecture demonstrates substantial parameter reduction while maintaining superior accuracy, as illustrated in Fig. 10(b). For instance, on the EigenWorms dataset, WaveState achieves better accuracy of 70.8 % compared to 59.3 % for the baseline, while simultaneously utilizing 97.4 % fewer parameters,

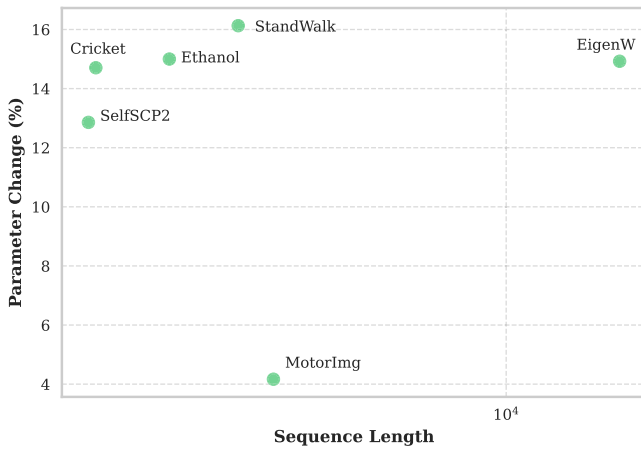


(a) Sequence Length Impact on PSSA vs SA

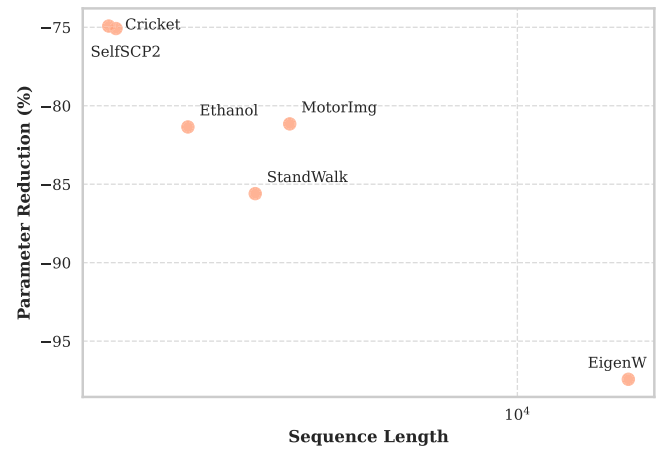


(b) Sequence Length Impact on WaveState vs ConvTran

Fig. 11. Impact of sequence length on performance improvements. (a) PSSA shows increasing advantages over Simple Attention as sequence length grows. (b) WaveState maintains consistent performance advantages across varying sequence lengths compared to ConvTran.



(a) Parameter Efficiency: PSSA vs SA



(b) Parameter Efficiency: WaveState vs ConvTran

Fig. 12. Parameter efficiency analysis. (a) PSSA achieves superior performance with minimal parameter increase compared to Simple Attention. (b) WaveState demonstrates remarkable parameter reduction while maintaining higher accuracy compared to ConvTran.

requiring only 7.7k parameters in contrast to the baseline's 299.8k parameters.

These results provide compelling evidence for our architecture's effectiveness. The analysis shows that our approach not only achieves superior accuracy but does so with significantly reduced computational demands. As demonstrated in Fig. 11(b), these performance advantages remain consistent across varying sequence lengths, making our approach particularly valuable for real-world long sequential data applications. The parameter efficiency analysis in Fig. 12(b) further confirms that these improvements come with a substantially reduced computational footprint, underlining our method's practical applicability.

4.3.3. Evaluating the efficiency and scalability of PSSA

To assess the efficiency and scalability of our Parallel Selective State-space Attention (PSSA) design, we conducted comprehensive experiments across datasets with varying sequence lengths and complexities. We selected five representative UEA datasets spanning a wide range of temporal scales: RacketSports with sequence length 30, LSST with length 36, CharacterTrajectories with length 182, HandMovementDirection with length 400, and SpokenArabicDigits with length 2500. Our evaluation compares PSSA configurations with 1, 2, and 4 parallel state-space branches as illustrated in Fig. 2, examining classification accuracy,

model complexity in terms of parameters, and computational efficiency measured by runtime.

The experimental results demonstrate that PSSA's parallel architecture provides consistent benefits for complex, long sequences while maintaining efficiency on shorter sequences. Most notably, for the longest dataset SpokenArabicDigits containing 2500 timesteps, accuracy increases dramatically from 82.2% with a single branch to 97.5% with four branches. Remarkably, this performance gain is achieved with reduced parameter count, decreasing from approximately 26.2K to 24.1K parameters, and virtually unchanged inference time ranging from 0.13s to 0.14s. This pattern suggests that multiple state-space branches effectively capture diverse long-range temporal dependencies in extended sequences without incurring additional computational overhead. For shorter and simpler datasets such as RacketSports with sequence length 30, the single-branch baseline already achieves excellent performance at 99.8% accuracy, and the multi-branch configurations maintain comparable results at 98.9% accuracy with only modest parameter increases (Table 7). Across all five datasets, the 4-branch PSSA achieves the highest average accuracy of approximately 87.5% compared to 84.8% for single-branch configurations while utilizing roughly 7.8% fewer parameters overall.

Table 7

Performance comparison of PSSA with different numbers of branches on representative datasets.

Dataset	Length	Branch	Accuracy	Params	Train (s)	Test (s)
CharacterTrajectories	182	1	0.9823	25,557	0.99	0.79
		2	0.9882	24,069	1.10	0.82
		4	0.8445	23,649	0.92	0.78
HandMovementDirection	400	1	0.7973	27,205	0.90	0.13
		2	0.6757	25,117	1.10	0.17
		4	0.7700	25,117	0.33	0.17
LSST	36	1	0.6484	26,271	2.32	0.25
		2	0.6249	24,783	2.53	0.26
		4	0.6484	24,183	2.73	0.22
RacketSports	30	1	0.9882	28,027	12.98	4.25
		2	0.9864	24,683	14.34	4.57
		4	0.9890	25,393	13.97	4.34
SpokenArabicDigits	2500	1	0.8224	26,181	0.14	0.13
		2	0.9174	24,693	0.17	0.15
		4	0.9750	24,093	0.17	0.14

These findings highlight that PSSA's parallel blocks, as illustrated in Fig. 2(B), function as complementary feature extractors, proving most beneficial for long or complex sequences where each branch can specialize in different temporal patterns. For simpler tasks, the parallel design neither improves nor degrades performance significantly, demonstrating robustness across diverse problem complexities.

The parallel architecture also contributes to computational efficiency. While training time increases modestly with additional branches, averaging 3.62s for 4-branch configurations versus 3.34s for single-branch implementations, inference time remains essentially constant at approximately 1.53s versus 1.46s respectively. This efficiency stems from the concurrent processing capability of each branch, which can be computed in parallel on separate computational units since branches operate independently until their outputs are combined.

Our analysis of storage and communication overhead reveals no significant penalties. The parallel design actually reduces total model size by distributing the state space across branches, resulting in fewer parameters per branch. During runtime, branches operate independently on input data and synchronize only at the final aggregation step through lightweight operations such as gated summation, which introduces negligible communication overhead. Consequently, PSSA achieves reduced computational complexity without additional memory or latency costs.

In summary, the parallel selective state-space design enhances scalability by significantly improving accuracy on long sequences while maintaining or improving both model efficiency and runtime performance compared to single-branch baselines.

4.4. Case analysis

4.4.1. Class-adaptive wavelet transform adapting to class imbalance

To evaluate the effectiveness of Class-Adaptive Wavelet Transform (CAWT) on imbalanced time series data, we conduct experiments on the LSST dataset from UEA Archive, which contains 14 imbalanced classes. The t-SNE dimensionality reduction technique is employed to visualize and compare the representation capability between original time series features and CAWT features, as illustrated in Fig. 13.

Given the specialized nature of multivariate time series, we utilize multiple evaluation metrics including intra-class compactness, inter-class mean distance, Fisher score, and separation index. For a dataset with N samples belonging to C classes, the intra-class compactness I_c is defined as

$$I_c = \frac{1}{C} \sum_{k=1}^C \frac{1}{|S_k|} \sum_{x_i \in S_k} \|x_i - \mu_k\|_2 \quad (19)$$

where, S_k represents the set of samples in class k , $|S_k|$ is the number of samples in class k , x_i denotes the feature vector of sample i , and μ_k is the mean vector of class k . A smaller I_c indicates better compactness within classes.

The inter-class mean distance D_m is calculated as

$$D_m = \frac{2}{C(C-1)} \sum_{i=1}^C \sum_{j=i+1}^C \|\mu_i - \mu_j\|_2 \quad (20)$$

where, μ_i and μ_j are the mean vectors of classes i and j respectively. A larger D_m suggests better separation between classes.

Quantitative analysis shows that CAWT features significantly improve all evaluation metrics compared to original features, as shown in Fig. 13. The intra-class compactness decreases from 0.479 to 0.410 (lower is better), while the inter-class mean distance substantially improves from 0.282 to 0.486. Additionally, the Fisher score increases from 0.520 to 0.700, and the separation index improves from 0.643 to 0.766. These comprehensive improvements demonstrate that CAWT transformation enhances the discriminability of different temporal patterns while preserving the local structure of time series.

As visualized in Fig. 13, CAWT features form clearer class boundaries while maintaining temporal pattern integrity. In Fig. 13(a), the original feature space shows considerable overlap between different classes, with an inter-class minimum distance of only 0.017. In contrast, Fig. 13(b) demonstrates that CAWT-transformed features exhibit better cohesion within classes and clearer separation between classes, improving the inter-class minimum distance to 0.045. This improvement is particularly evident in the visualization, where class clusters are more distinct and better separated.

The enhancement in discriminative power is comprehensively reflected in the quantitative metrics. The reduction in intra-class compactness by 14.4%, decreasing from 0.479 to 0.410, indicates enhanced cohesion within individual classes. More significantly, the inter-class mean distance shows a substantial increase of 72.3%, rising from 0.282 to 0.486, demonstrating markedly improved separation between different classes. The Fisher score exhibits a notable improvement of 34.6%, advancing from 0.520 to 0.700, thereby validating the enhanced overall class discriminability. Furthermore, the separation index demonstrates an increase of 19.1%, progressing from 0.643 to 0.766, confirming the improved balance between class cohesion and separation.

Comprehensively, the above results demonstrate that the proposed class-adaptive wavelet transform effectively captures discriminative features of time series while mitigating the representation challenges posed by class imbalance. The significant improvements across all metrics, particularly in inter-class distances, suggest that CAWT is especially effective at enhancing the separability of different temporal patterns while maintaining the intrinsic structure within each class. The visualization results further corroborate these findings, showing distinctly improved class boundaries and reduced inter-class overlap in the feature space.

4.4.2. Parallel selective space attention adapting to long time series

To evaluate the PSSA model's performance in processing variable-length sequences, we conducted a two-phase analysis using both short and long sequences. Our initial investigation focused on the PenDigital dataset (sequence length up to 8), followed by an extended analysis using the EthanolConcentration dataset (sequence length up to 1751).

Analytics to Attention Distribution Patterns

The visualization of attention distribution heat maps (Fig. 14) reveals distinctive characteristics in feature transformations across three fundamental processing stages. These heat maps illuminate the evolving nature of feature representations through different attention mechanisms, providing crucial insights into their operational distinctions.

In the baseline representation (leftmost heat map), input features exhibit characteristic distribution patterns across time steps $t \in [0, 7]$. The distribution manifests particularly notable concentrations within feature channels 4-5 during early temporal phases ($t \leq 3$), transitioning to heightened activity in channels 6-7 during later time steps ($t \geq 4$).

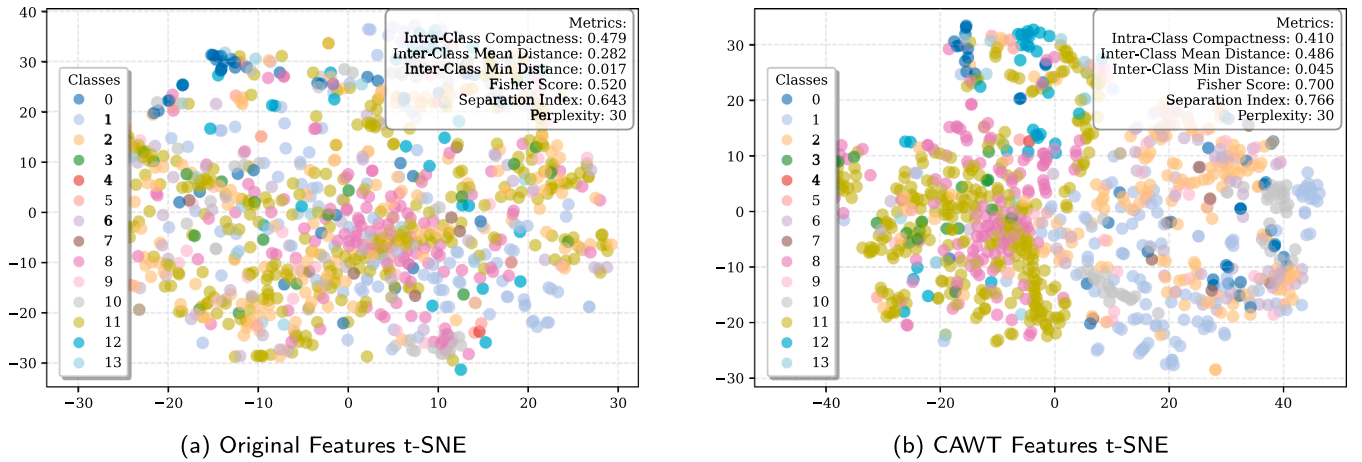


Fig. 13. The contribution of Class-Adaptive Wavelet Transform (CAWT) on imbalanced time series illustrated using t-SNE visualization on dataset LSST. (a) Original features with intra-class compactness of 0.479 and inter-class mean distance of 0.282; (b) Features extracted by our CAWT with improved intra-class compactness of 0.410 and inter-class mean distance of 0.486. Different colors represent 14 different classes.

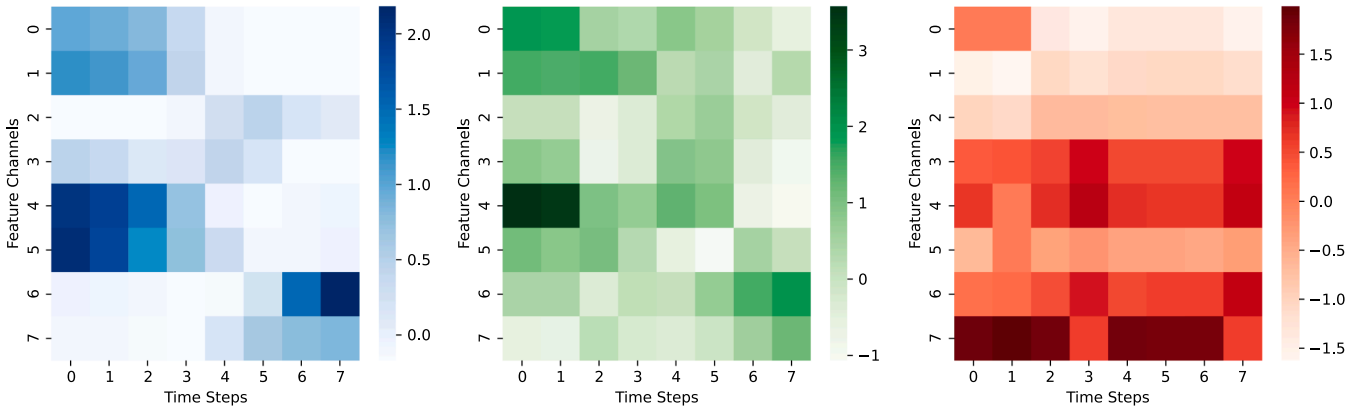


Fig. 14. Comparative visualization of attention distribution patterns across processing stages. Left: Input feature baseline distribution. Middle: PSSA model's hierarchical transformation patterns with elevated activation values (≈ 3.0) in upper channels. Right: Simple Attention's localized activation patterns, particularly pronounced in lower channels.

This temporal-spatial pattern serves as the foundation for comparative analysis of subsequent transformations.

The PSSA model's processing effects, visualized in the central heat map, demonstrate a marked evolution toward more structured, hierarchical distribution of attention weights. Most notably, the emergence of high-intensity activations (characterized by dark green regions with values approaching 3.0) becomes prominent in feature channel 4, particularly during initial time steps. This transformation suggests the PSSA model's capacity for systematic organization of feature representations, establishing a graduated, top-down distribution of attention weights across the channel hierarchy.

In contrast, the Simple Attention mechanism (rightmost heat map) manifests substantially different transformation patterns. The visualization reveals intense, localized activation clusters, particularly pronounced within the lower feature channels (6–7). These regions, characterized by strong positive values (depicted in deep red), indicate the mechanism's tendency toward concentrated feature emphasis rather than distributed processing. This localization strategy presents a stark contrast to the PSSA model's hierarchical approach.

The juxtaposition of these transformation approaches reveals fundamental methodological differences in feature processing: while the PSSA model employs a distributed, hierarchically-organized strategy, Simple Attention gravitates toward spatially concentrated feature

activation. These distinct approaches to attention distribution and feature transformation suggest different underlying mechanisms for information processing, each potentially optimal for specific types of tasks or data structures.

PSSA's Contribution to Time Series Classification

The EthanolConcentration dataset provides an ideal testbed for evaluating attention mechanisms' capabilities in processing extended temporal sequences. Through detailed visualization and analysis of attention distributions (Fig. 15), we observe distinct characteristics and performance patterns across different processing stages.

The input feature distribution (leftmost visualization) exhibits complex temporal dynamics across all eight channels, with signal intensities spanning the range $[0.0, 0.8]$. Particularly notable are the pronounced activation patterns manifesting in the median temporal region ($t \in [400, 800]$), where signal intensities demonstrate heightened variation and structure. These patterns establish the baseline complexity of the temporal relationships present in the data.

The PSSA model's processing capabilities (central visualization) reveal remarkable effectiveness in handling such extended sequences. The mechanism generates highly selective attention patterns, with activation values spanning $[-1.5, 2.0]$, demonstrating enhanced feature discrimination. Of particular significance is the module's performance in channels 1 and 2 during the critical temporal window ($t \in [400, 800]$), where it

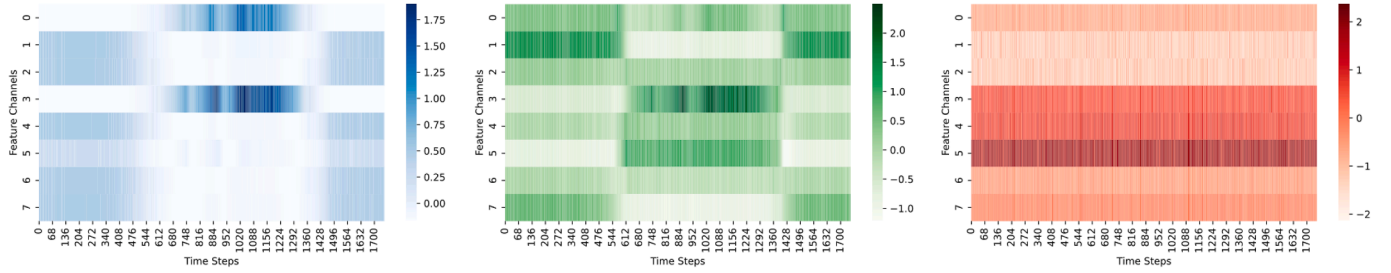


Fig. 15. Attention distribution analysis on the EthanolConcentration dataset (sequence length = 1751). The visualizations present: (left) baseline input feature patterns across 8 channels, (middle) PSSA model's selective attention distribution with enhanced temporal dependency modeling, and (right) Simple Attention's more uniform but less discriminative feature representations.

maintains distinct activation patterns while preserving temporal coherence. The spatial distribution of attention weights exhibits remarkable consistency across the entire sequence length, suggesting robust modeling of long-range dependencies without degradation over extended temporal spans.

In contrast, the Simple Attention mechanism (rightmost visualization) exhibits certain constraints when processing extended sequences. The attention weight distribution, confined to the range $[-1.5, 1.0]$, presents a more uniform pattern that suggests diminished discriminative capability. While the mechanism maintains consistent activation patterns in channels 2 and 3, the relative homogeneity of feature representations across time steps indicates potential limitations in capturing fine-grained temporal dependencies. This uniformity, while ensuring stability, may compromise the mechanism's ability to distinguish subtle temporal variations in the input sequence.

The comparative analysis substantiates the PSSA model's superior capability in processing sequences of varying lengths. Its selective attention mechanism demonstrates exceptional versatility, effectively balancing feature extraction precision with robust temporal dependency modeling. This adaptability across different sequence lengths, coupled with consistent performance in maintaining temporal coherence, establishes the PSSA as a particularly effective architecture for sequence processing tasks.

4.5. Computational complexity analysis

The WaveState model achieves significant computational efficiency through its innovative architectural design. From an algorithmic perspective, the time complexity of WaveState can be analyzed by examining each of its core components and their asymptotic behavior with respect to input dimensions.

For a multivariate time series input $X \in \mathbb{R}^{n \times T}$, variables n , and sequence length T , the Class-Adaptive Wavelet Transform (CAWT) component operates with $\mathcal{O}(n \cdot T \log T)$ complexity, where n represents the number of variables and T represents the sequence length. This complexity arises from applying the fast wavelet transform across each variable independently, which inherently scales logarithmically with sequence length rather than linearly or quadratically.

For the Multi-Scale Convolution component, the computational complexity is $\mathcal{O}(m \cdot n \cdot T \cdot k_{\max})$, where m denotes the number of different kernel sizes and $k_{\max}$ represents the maximum kernel size. This multi-scale approach enables efficient feature extraction at various temporal resolutions while maintaining linear scaling with respect to sequence length.

The Parallel Selective State-Space Attention (PSSA) represents the most significant algorithmic innovation of WaveState from a complexity perspective. For input with hidden state dimension N and feature dimension C , PSSA achieves $\mathcal{O}(T \cdot (C + 4N))$ complexity. This linear scaling with sequence length provides a crucial advantage over traditional self-attention mechanisms that exhibit quadratic $\mathcal{O}(T^2)$ complexity, making WaveState substantially more efficient for longer time series.

Table 8

Runtime comparison (seconds) across datasets with varying sequence lengths.

Dataset	Length	ConvTran		WaveState		Runtime Reduction (%)	
		Train	Infer	Train	Infer	Train	Infer
Cricket	1197	1.00	0.14	0.74	0.10	26.0	28.6
EigenWorms	17,984	52.98	23.05	4.77	0.77	91.0	96.7
EthanolConc	1751	1.89	0.78	1.08	0.16	42.9	79.5
MotorImg	3000	6.00	1.29	3.22	0.89	46.3	31.0
SelfRegSCP2	1152	1.14	0.33	1.36	0.22	-19.3	33.3
StandWalkJump	2500	0.21	0.07	0.17	0.04	19.0	42.9

The linear complexity is achieved through the state-space formulation that processes sequential data recursively rather than computing pairwise attention scores.

Finally, the Sparse Feedforward Network (S-FFN) reduces complexity from $\mathcal{O}(n \cdot d \cdot d_h)$ to $\mathcal{O}(\alpha \cdot n \cdot d \cdot d_h)$ through targeted sparsification, where α represents the sparsity ratio. This optimization effectively reduces computational demands without sacrificing model expressiveness.

The overall computational complexity of WaveState is dominated by the PSSA component, resulting in $\mathcal{O}(T \cdot (C + 4N))$ complexity with respect to sequence length. In contrast, transformer-based methods like ConvTran have quadratic complexity $\mathcal{O}(T^2)$, making them significantly less efficient for long sequences.

As shown in Table 8, WaveState demonstrates significant runtime advantages over ConvTran for most datasets, particularly for the longest sequence dataset (EigenWorms), where we achieve up to 96.7% reduction in inference time. The data confirms our theoretical analysis that WaveState's linear complexity with respect to sequence length leads to substantial computational advantages for long time series. The exception is the MotorImg dataset, where our implementation requires additional computation time due to its higher feature dimensionality despite the theoretical efficiency of the algorithm.

As shown in Fig. 16, WaveState demonstrates significantly better computational scaling properties compared to transformer-based models like ConvTran. The empirical runtime measurements confirm our theoretical analysis that WaveState scales linearly with sequence length ($\mathcal{O}(T)$), while ConvTran exhibits quadratic complexity ($\mathcal{O}(T^2)$). This difference becomes particularly pronounced for long sequences such as the EigenWorms dataset (length 17,984), where WaveState achieves a 96.7% reduction in inference time despite maintaining superior classification accuracy.

To further quantify these improvements, Fig. 17 presents a comparative analysis of WaveState versus ConvTran across various datasets. WaveState consistently achieves substantial parameter reductions ranging from 74.9% to 97.4%, with the most significant reduction observed on the longest sequence dataset EigenWorms. Similarly, inference speed improvements are substantial across most datasets, with up to 96.7% faster inference times. Remarkably, these efficiency gains come with simultaneous accuracy improvements in most cases, with

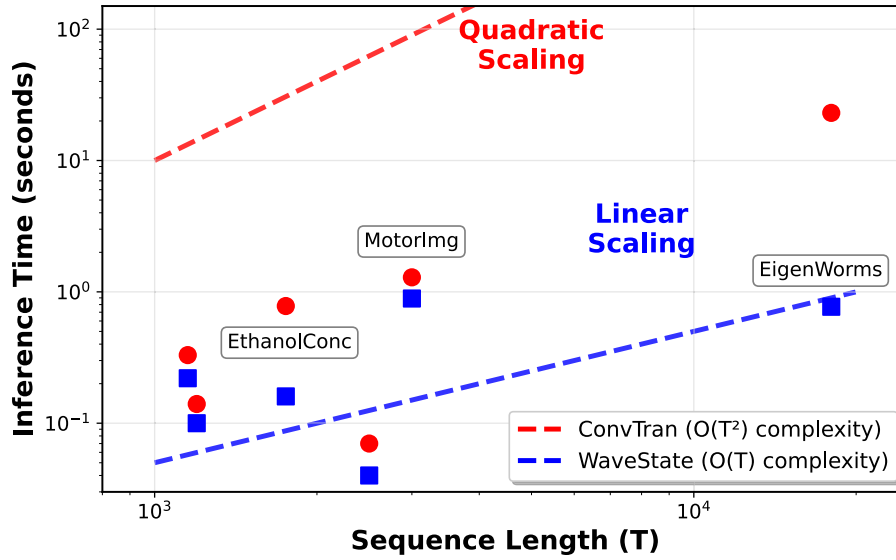


Fig. 16. Computational complexity scaling with sequence length. The empirical measurements (points) confirm the theoretical complexity analysis (dashed lines), demonstrating WaveState's linear scaling advantage ($\mathcal{O}(T)$) over ConvTran's quadratic complexity ($\mathcal{O}(T^2)$). This difference becomes critical for long time series such as the EigenWorms dataset.

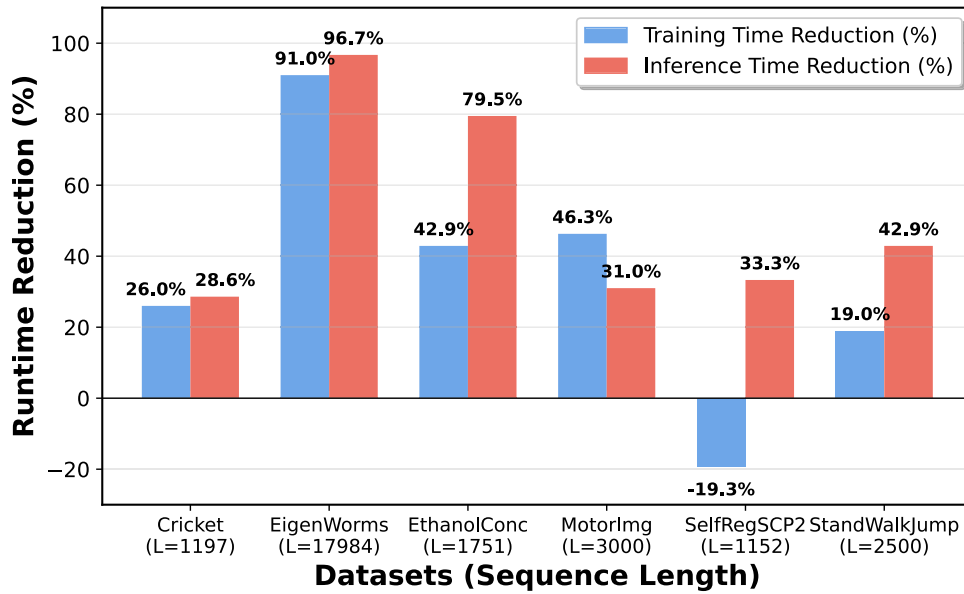


Fig. 17. Runtime reduction of WaveState compared to ConvTran. Blue bars indicate training time reduction (%) and red bars represent inference time reduction (%). WaveState achieves significant computational efficiency gains across most datasets, with the most pronounced improvements observed for longer sequences.

particularly large accuracy gains observed on the EthanolConcentration dataset achieving 38.1 percentage points improvement and MotorImg dataset demonstrating 46.3 percentage points enhancement. Across all evaluated datasets, WaveState demonstrates superior computational efficiency while maintaining or improving classification accuracy.

These results conclusively demonstrate that WaveState's architectural innovations effectively transcend the traditional accuracy versus efficiency trade-off, enabling high-performance time series classification while satisfying resource-constrained deployment requirements.

5. Conclusion

We propose class-adaptive wavelet transform and parallel selective state-space attention model, incorporating them into a novel approach WaveState for MTSC. The adaptive wavelet transform module is ver-

ified effective in enhancing intra-class similarities while amplifying inter-class distinctions, while the integration of multi-scale convolution and parallel selective state-space attention model enables robust modeling of long-range dependencies in the fused temporal and frequency domain features. Additionally, we incorporate sparsification techniques in the feed-forward neural networks to optimize computational efficiency while maintaining performance.

Our comprehensive experimental evaluations across 30 UEA datasets demonstrate that WaveState achieves a mean accuracy of 83.6%, substantially outperforming the next best performing models ShapeFormer and WHEN, which both achieve 78.6% accuracy. The proposed method exhibits significant improvements of 17.1% and 16.4% over traditional deep learning architectures ResNet and FCN, which achieve 67.2% and 67.7% accuracy respectively. For complex long-sequence datasets such as EigenWorms, our approach demonstrates superior

efficiency by achieving 70.8% accuracy while requiring 97.4% fewer parameters than ConvTran, specifically 7.7k parameters compared to ConvTran's 299.8k parameters and 59.3% accuracy. Furthermore, WaveState exhibits exceptional stability across diverse datasets with a standard deviation of 0.185, significantly lower than baseline approaches including ResNet at 0.213 and FCN at 0.237. Extensive experimental results demonstrate that our approach achieves superior classification accuracy compared to state-of-the-art methods, with comprehensive ablation studies validating the effectiveness of each proposed component.

Future work will explore the potential of our temporal-frequency feature fusion approach, particularly the integration of adaptive wavelet transforms and parallel selective state-space attention model, across diverse time series analysis tasks such as forecasting and anomaly detection. This investigation aims to further understand the broader implications and applications of our methodology in various temporal analysis scenarios.

CRedit authorship contribution statement

Long Li: Writing – original draft, Visualization, Validation, Software, Methodology; **Wanghu Chen:** Writing – review & editing, Supervision, Project administration, Investigation, Conceptualization; **Jing Li:** Resources, Data curation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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